PROBLEM 04

Plane at Constant Altitude

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Coordinate Transformations

An aircraft moves at constant altitude y_0 with velocity v_0 in the x direction. At time $t = 0$ it is located at $x = 0$, and it begins to accelerate in the x direction at a constant rate a_x. Describe the position, velocity, and acceleration in polar coordinates r, θ, and answer the following questions:

(a) Show that: $\dot{r} = \frac{v(t)x(t)}{r(t)}$

(b) Show that: $\dot{\theta} = -\frac{v(t)y_0}{r^2(t)}$

(c) Show that: $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r^3(t)}$

(d) Show that: $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r^4(t)}$

From these expressions form the acceleration vector $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$. Repeat the problem using Cartesian coordinates.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An aircraft flies at constant altitude y_0 with velocity in the x direction. At $t = 0$, $x(0) = 0$ and velocity is v_0. It accelerates in the x direction at constant rate a_x.
- **Tracking System:** A ground radar station located at origin $O(0, 0)$ tracks the aircraft using range $r(t)$ and elevation/bearing angle $\theta(t)$.
- **Hypotheses:** $y_0 > 0, v_0 \geq 0, a_x = \text{const}$. Altitude is fixed: $y(t) \equiv y_0 \implies \dot{y} = 0, \ddot{y} = 0$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** Coordinates (x, y) with position $\mathbf{r} = x\mathbf{i} + y_0\mathbf{j}$.
- **Radar Polar Frame:** Position $\mathbf{r} = r\mathbf{e}_r$, where:

$$r(t) = \sqrt{x(t)^2 + y_0^2}, \quad \tan\theta(t) = \frac{y_0}{x(t)} \implies \theta(t) = \arctan\left(\frac{y_0}{x(t)}\right)$$

- **Kinematics in x:**

$$\dot{x}(t) = v(t) = v_0 + a_x t, \quad \ddot{x}(t) = a_x$$

$$x(t) = v_0 t + \frac{1}{2} a_x t^2$$

- **Change of basis (radar polar basis).** With θ the angle of the line of sight measured from $+\mathbf{i}$, the polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$. The columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}$ in $\mathcal{B}_0$; using $\cos\theta = x/r$ and

$\sin\theta = y_0/r$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x/r & -y_0/r & 0 \\ y_0/r & x/r & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det[{}_0R_1] = \frac{x^2 + y_0^2}{r^2} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Both properties were verified symbolically with sympy. The inverse relation $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$ (first row of $[{}_0R_1]^T$ applied to the basis) is the one used in the Cartesian cross-check of Phase 3.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Proof that $\dot{r} = \frac{v(t)x(t)}{r(t)}$

Pedagogical Justification: Starting from the geometric relation $r(t)^2 = x(t)^2 + y_0^2$, we differentiate implicitly with respect to time t.

- **Implicit Differentiation:**

$$\frac{d}{dt}[r^2] = \frac{d}{dt}[x^2 + y_0^2]$$

Applying the chain rule $\frac{d}{dt}(r^2) = 2r\dot{r}$ and $\frac{d}{dt}(x^2) = 2x\dot{x}$:

$$2r\dot{r} = 2x\dot{x} + 0 = 2xv(t)$$

Dividing by $2r$:

$$\dot{r} = \frac{v(t)x(t)}{r(t)} \quad \blacksquare$$

(b) Proof that $\dot{\theta} = -\frac{v(t)y_0}{r(t)^2}$

> **Pedagogical Justification:** Starting from $\tan\theta = \frac{y_0}{x}$, we differentiate both sides with respect to time using the quotient rule.

- **Differentiation:**

$$\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{y_0}{x}\right)$$

Recalling $\frac{d}{dt}(\tan\theta) = (1+\tan^2\theta)\dot{\theta} = \sec^2\theta\,\dot{\theta}$, and $\frac{d}{dt}(x^{-1}) = -\frac{\dot{x}}{x^2} = -\frac{v}{x^2}$:

$$\sec^2\theta\,\dot{\theta} = -\frac{y_0 v(t)}{x(t)^2}$$

From geometry: $\cos\theta = \frac{x}{r} \implies \sec\theta = \frac{r}{x} \implies \sec^2\theta = \frac{r^2}{x^2}$:

$$\frac{r^2}{x^2}\dot{\theta} = -\frac{y_0 v(t)}{x^2}$$

Multiplying both sides by x^2 and dividing by r^2:

$$\dot{\theta} = -\frac{v(t)y_0}{r(t)^2} \quad \blacksquare$$

(c) Proof that $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3}$

> **Pedagogical Justification:** We differentiate the expression for $\dot{r} = \frac{vx}{r}$ using the quotient rule $\left(\frac{u}{w}\right)' = \frac{u'w - uw'}{w^2}$.

- **Differentiation:**

Here $u = vx \implies \dot{u} = \dot{v}x + v\dot{x} = a_x x + v^2$.

The denominator is $w = r \implies \dot{w} = \dot{r} = \frac{vx}{r}$.

$$\ddot{r} = \frac{\dot{u}r - u\dot{r}}{r^2} = \frac{(a_x x + v^2)r - (vx)\left(\frac{vx}{r}\right)}{r^2}$$

Multiplying numerator and denominator by r:

$$\ddot{r} = \frac{(a_x x + v^2)r^2 - v^2 x^2}{r^3}$$

Expanding the numerator:

$$(a_x x + v^2)r^2 - v^2 x^2 = a_x x r^2 + v^2 r^2 - v^2 x^2 = a_x x r^2 + v^2(r^2 - x^2)$$

Recalling that $r^2 - x^2 = y_0^2$:

$$\ddot{r} = \frac{a_x x(t) r^2(t) + v^2(t) y_0^2}{r(t)^3} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3} \quad \blacksquare$$

(d) Proof that $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4}$

> **Pedagogical Justification:** We differentiate $\dot{\theta} = -y_0 v r^{-2}$ with respect to time using the product rule.

- **Differentiation:**

$$\ddot{\theta} = \frac{d}{dt}\left(-y_0 \frac{v}{r^2}\right) = -y_0 \frac{\dot{v}r^2 - v\frac{d}{dt}(r^2)}{r^4}$$

Substituting $\dot{v} = a_x$ and $\frac{d}{dt}(r^2) = 2r\dot{r} = 2r\left(\frac{vx}{r}\right) = 2vx$:

$$\ddot{\theta} = -y_0 \frac{a_x r^2 - v(2vx)}{r^4} = -y_0 \frac{a_x r^2 - 2v^2 x}{r^4} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4} \quad \blacksquare$$

Polar and Cartesian Acceleration Vectors

- **Polar Acceleration Vector:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - r\left(-\frac{v y_0}{r^2}\right)^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - \frac{v^2 y_0^2}{r^3} = \frac{a_x x r^2}{r^3} = a_x \frac{x}{r}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\left(y_0 \frac{2v^2 x - a_x r^2}{r^4}\right) + 2\left(\frac{vx}{r}\right)\left(-\frac{v y_0}{r^2}\right) = \frac{2v^2 x y_0 - a_x y_0 r^2}{r^3} - \frac{2v^2 x y_0}{r^3} = -a_x \frac{y_0}{r}$$

$$\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$$

- **Cartesian Verification:**

In Cartesian coordinates:

$$\mathbf{a} = \ddot{x}\mathbf{i}_0 + \ddot{y}\mathbf{j}_0 = a_x\mathbf{i}_0 + 0\mathbf{j}_0 = a_x\mathbf{i}_0$$

Converting Cartesian $\mathbf{i}_0$ into polar: $\mathbf{i}_0 = \cos\theta\mathbf{e}_r - \sin\theta\mathbf{e}_\theta = \frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta$.

Substituting gives identically $\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$. The match is mathematically exact!

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:**
- $[\dot{r}] = [v\,x/r] = \mathrm{m/s}$; $\quad [\dot{\theta}] = [v\,y_0/r^2] = \dfrac{(\mathrm{m/s})\,\mathrm{m}}{\mathrm{m}^2} = \mathrm{s}^{-1}$ (rad/s).
- $[\ddot{r}] = \dfrac{(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$ and $\dfrac{(\mathrm{m/s}^2)\,\mathrm{m}\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$; $\quad [\ddot{\theta}] = \dfrac{\mathrm{m}\,(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}}{\mathrm{m}^4} = \mathrm{s}^{-2}$ (Consistent).
- **Limit $x \to \infty$ (aircraft far away):** $r \to x$, so $\dot{r} \to v$, $\dot{\theta} \to 0$, $a_r \to a_x$, $a_\theta \to 0$: the line of sight aligns with the flight direction and the motion is purely radial.
- **Limit $x = 0$ (aircraft overhead, $t = 0$):** $\dot{r} = 0$, $\dot{\theta} = -v_0/y_0$, $a_r = 0$, $a_\theta = -a_x$. Then $\theta = \pi/2$ and $\mathbf{a} = a_x\mathbf{i} = -a_x\mathbf{e}_\theta$, as required by the change of basis.
- **Invariance of the modulus:** $a_r^2 + a_\theta^2 = a_x^2\,\dfrac{x^2 + y_0^2}{r^2} = a_x^2$, equal to $\|a_x\mathbf{i}\|^2$ because $[{}_0R_1]$ is orthogonal.
- **Symbolic check (sympy):** differentiating $r = \sqrt{x^2 + y_0^2}$ and $\theta = \arctan(y_0/x)$ with $x = v_0 t + \frac{1}{2}a_x t^2$ reproduces the four identities (a)-(d) with zero residual, and gives $a_r = a_x x/r$, $a_\theta = -a_x y_0/r$ with zero residual; the three limits above were obtained with sympy as well.

Source: Problems.pdf